Wave speed

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use speed, frequency and wavelength in a consistent relationship.

SUCCESS CRITERIA - I can...

1. Use v=fλ with matching units.
2. Check frequency and wavelength changes at fixed speed.
3. Distinguish medium speed from oscillation rate.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A heat engine operates between 500 K and 300 K and absorbs 700 J. Find maximum ideal work and explain why actual work is usually lower.

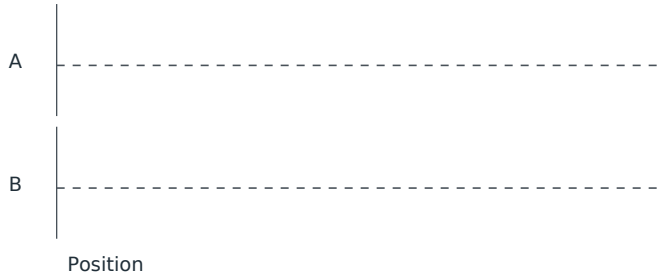
Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: v = f λ; λ = v/f.
At fixed wave speed v, increasing frequency f decreases wavelength λ.

Worked reasoning: A wave with wavelength 3 m and frequency 2 Hz travels at v=f λ=6 m/s. At the same speed, frequency 3 Hz would require wavelength 2 m.

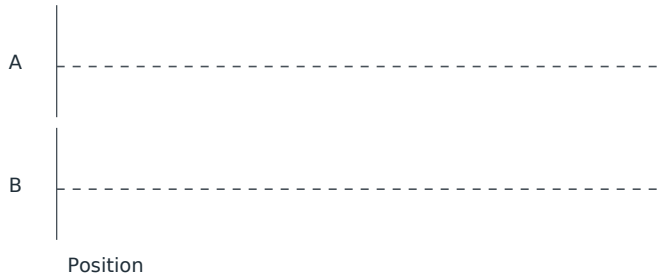
Your turn: Draw both waves over the same 6 m length and label their frequencies.



Marzano check - circle: ① ② ③

03 Read and interpret the model

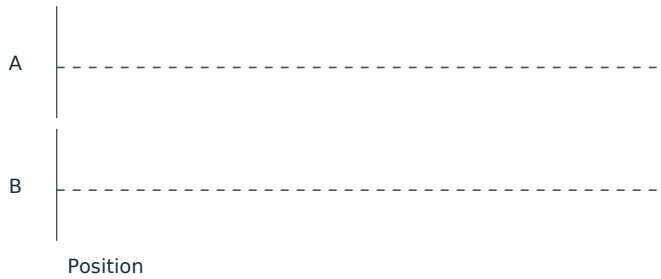
Two waves travel in the same ideal medium at the same speed. One has twice the frequency. Compare wavelengths.



Marzano check - circle: ① ② ③

04 Apply the physics

A speaker produces a 425 Hz tone in air where sound speed is 340 m/s. Calculate its wavelength. Find the frequency that would produce a 0.50 m wavelength in the same air. Sketch the relative crest spacings.



Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

A student writes v=f/λ. Check units and explain why that is not a speed.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

An object at 360 K sits in a vacuum facing surroundings at 240 K. How can it cool, and are absolute-zero surroundings required?
Label temperatures, samples and relevant energy transfers.

Sample A

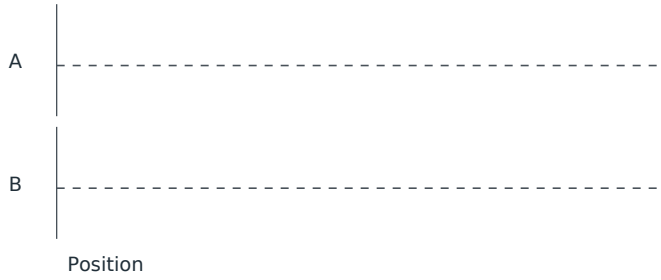
Sample B

Label temperatures and draw energy-transfer arrows.

Marzano check - circle: 1 2 3

07 Exit check

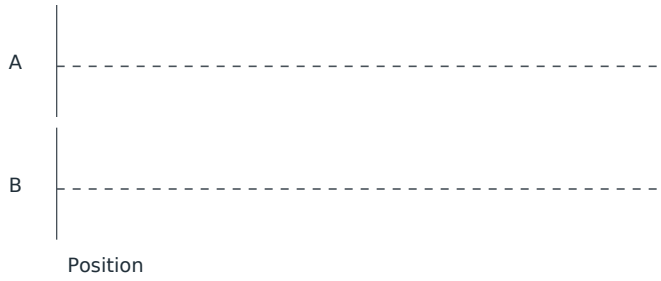
A wave travels 12 m in 0.40 s and has frequency 50 Hz. Find speed and wavelength 0.



Marzano check - circle: 1 2 3

08 STRETCH: test the model

A wave enters a medium where speed changes from 12 to 8 m/s. Frequency stays 4 Hz. Find both wavelengths and explain why frequency does not reset at the boundary.



Marzano check - circle: 1 2 3